RAJKUMAR VIJAYAN

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LinkedIn | GitHub | Portfolio

PROFESSIONAL PROFILE

MSc Software Development candidate (May 2027) with hands-on C++ and C++20 concurrency, thread-safe caching, priority-based scheduling and latency measurement projects. Experience developing Java/Spring Boot APIs, automated tests and data-quality workflows. Working knowledge of Go and CI/CD from projects. Available for onsite software engineering work in Dublin from July 2027.

TECHNICAL SKILLS

Languages: C++, C++20, Go, Java, Python, SQL, C#, JavaScript

Systems and algorithms: Data structures, hash maps, linked lists, priority queues, multithreading, mutexes, condition variables, thread pools, dynamic batching

Engineering: Object-oriented design, debugging, issue investigation, REST APIs, automated testing, throughput and p50/p95/p99 latency metrics

Tools: Linux/WSL, Git/GitHub, CI/CD, Docker, CMake, GoogleTest, JUnit, PIT, Spring Boot, PostgreSQL, Azure

SOFTWARE ENGINEERING PROJECTS

InferX — Concurrent AI Inference Scheduler

C++20 | Thread Pools | Performance Measurement

- Developed a concurrent scheduler using dynamic batching, a priority heap, worker threads and condition-variable coordination to dispatch tasks.
- Implemented priority-based execution and measured throughput and p50/p95/p99 latency to evaluate scheduling behaviour and performance trade-offs.
- Applied synchronization primitives to coordinate worker execution and investigate task-scheduling behaviour.

CacheForge — Thread-Safe LRU/TTL Cache

C++20 | CMake | GoogleTest

- Built a reusable cache library and command-line interface using a hash map and linked list to support least-recently-used eviction and time-to-live expiration.
- Synchronized shared cache operations for concurrent access and wrote GoogleTest tests for cache operations and edge cases; all implemented tests passed.
- Structured the library around explicit cache operations and eviction behaviour to make the implementation testable and maintainable.

TradeMatch Exchange — Order Matching Backend

Java | Spring Boot | PostgreSQL

- Implemented buy/sell order matching using price-time priority and separated matching logic from API and database interactions.
- Developed REST endpoints, PostgreSQL persistence and CSV export for reviewing order activity.

TestForge / Dissertation Pilot — Mutation-Based Test Evaluation
Java | Python | JUnit | PIT

- Configured a reproducible Apache Commons Lang 3.9 pilot using JUnit and PIT mutation testing to evaluate Java test-suite effectiveness.
- Executed tests, generated mutation reports and inspected surviving mutants to identify test-coverage gaps; broader LLM benchmarking remains ongoing.

PROFESSIONAL EXPERIENCE

Amazon Development Centre India — ML Data Associate Feb 2022 – Jan 2023

- Validated more than 100,000 machine-learning data records, investigated anomalies and applied quality-control procedures to support reliable downstream datasets.
- Contributed to operational workflow improvements associated with an approximately 30% increase in team productivity.
- Documented data-quality findings and supported consistent review processes across the team.

VRBB & Associates — Business Consulting Intern May 2024 – Jul 2024

- Reviewed incorporation, financial and contractual records for due diligence; analysed operational intake and prepared structured findings and management reports.

EDUCATION

University of Limerick, Ireland Sep 2025 – May 2027 (expected)

MSc Software Development (International Systems)

- Dissertation: An Empirical Evaluation of Large Language Models for Automated Software Test Generation Using Mutation Testing.
- Relevant modules: Data Structures and Algorithms; Software Testing and Inspection; Text Analytics and NLP; Mobile Applications Development.

Thiagarajar School of Management, India 2023 – 2025

Postgraduate Diploma in Management (Business Analytics), First Class

B.S. Abdur Rahman Crescent Institute, India 2017 – 2020

Bachelor of Computer Applications (Data Science), First Class

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Entrepreneurship Cell — Core Committee Member (2023–2025): Coordinated approximately 15–20 sponsors and supported event technology and promotions for initiatives attended by over 1,000 people.

Certifications: Google Cybersecurity Professional Certificate; Linux Fundamentals; Google Data Analytics Professional Certificate; AWS Cloud Foundations.

ADDITIONAL INFORMATION

Availability: July 2027; open to relocation to Dublin.

Work authorisation: Currently on Stamp 2 student permission; intends to apply for Stamp 1G following graduation, subject to eligibility and approval. May require employment-permit support in the future.